MARIAGIULIA MARTINELLI



Bioarchaeology | Palaeogenetics | Conservation Science

mariagiuliamartinelli.github.io
mariagiulia.martinelli@unifi.it
mariagiulia-martinelli-19977b277 mariagiuliamartinelli 0009-0005-7332-8618

PROFILE

I am a PhD student in **Evolutionary Biology and Ecology** at the Universities of Florence and Parma, where I am beginning to work on the microorganisms associated with alpine plants and soils and on how these communities respond to climate change. I come to it from **bioarchaeology**, **palaeogenetics** and **conservation science**, where I studied what happens to biological material after deposition: the taphonomy of human skeletal remains, the microbial biodeterioration of stone and the survival of ancient biomolecules. Across these fields I work with the same tools: amplicon metagenomics, ancient DNA, and reproducible, openly released R pipelines run on HPC clusters.

EDUCATION

PhD in *Evolutionary Biology and Ecology* (XLII cycle), Curriculum in Ecology and Ethology

University of Florence / University of Parma

11/2026 – present

Master's Degree in *Science and Technology for the Conservation of Cultural Heritage* (LM-11), 110/110 cum laude

***Sapienza* University of Rome**

09/2024 – 10/2026

- Thesis: *Conservation and Reassessment of an Overlooked Skeletal Collection: Historical and Bioarchaeological Study of Human Crania from Etruscan Territory Acquired by G. Sergi in the Early 20th Century*.
- Erasmus+ semester at Stockholm University (01–06/2026): *Palaeogenetics*, *Modelling Tools for Environmental Scientific Studies*.

Bachelor's Degree in *Technologies for the Conservation and Restoration of Cultural Heritage* (L-43), 110/110

***Sapienza* University of Rome**

09/2021 – 12/2024

- Thesis: microbiota succession on the walls of the Casino Algardi (Rome) after conservation treatments with sustainable products.

RESEARCH EXPERIENCE

Research Internship, *Centre for Palaeogenetics*

Stockholm University

04/2026 – 06/2026

- Metagenomic classification (Kraken2, KrakenUniq) of 250 ancient DNA samples from Roman, Visigothic, Islamic and Medieval necropolises in Valencia, run as SLURM job arrays on the Dardel HPC cluster; cross-tool validation and pathogen screening.

Osteological Laboratory, *Tusculum* Project

EEHAR-CSIC, Rome

12/2025 – 01/2026

- Inventory, biological profile and taphonomic recording of medieval human remains; digitisation of the anthropological dataset.

Research Internship, *Palaeoanthropology and Bioarchaeology Laboratory*

Sapienza University of Rome

07/2025 – 12/2025

- Bioanthropological re-evaluation of the Sergi cranial collection: biological profile, dental palaeopathology and state of conservation, with a reproducible R pipeline.

Research Internship, ENEA Casaccia Research Centre

ENEA, Rome

04/2024 – 12/2024

- End-to-end amplicon metagenomics of biofilms on historic walls: sampling, DNA extraction, PCR (16S rRNA, ITS), sequencing and R analysis of diversity under sustainable treatments.

PUBLICATIONS AND TALKS

C. Alisi, V. Fonti, M. Martinelli, G. Migliore, P. Paganin, F. Tasso
Abstract, Proceedings of *The Omics & Heritage*, Rome, 2024
Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights

M. Martinelli, V. Fonti, P. Paganin, F. Tasso, C. Alisi
International Biodeterioration & Biodegradation, in preparation
Microbial Biodiversity on the Casino del Bel Respiro Walls: Implications for Conservation Strategies

L. Morucci, A.A. Bastanza, M. Martinelli, S. Iorio
International Journal of Paleopathology, in preparation
A Systematic Review of Analytical Methodologies in Bioarchaeology of Stress in Non-Adult Populations

Oral contribution, *Orizzonti Antichi*, University of Pisa, 12/2025
Glycyrrhiza glabra vs Biofilm: nuove prospettive per la conservazione dei beni culturali

AWARDS

Tutoring Fellowship, Degree Programmes in Applied Sciences for Cultural Heritage

Sapienza University of Rome

a.y. 2026/2027

Merit Scholarship, Academic Excellence Track (Bachelor's and Master's)

Sapienza University of Rome

2023 – 2026

LANGUAGES

Italian
Native speaker

English
B2 (CEFR)

Full CV and references available on request.